METHODOLOGY RECAP · MODULE M2

Data quality adjustment

What the module does · How to read its outputs

WHAT IT DOES

This module **repairs** the problems the previous one found. It fills the missing months and replaces the outlier spikes, so that trends and coverage aren't thrown off by a few bad numbers.

WHAT IT DOES NOT DO

It never invents a trend. Every replacement comes from the facility's **own history**, and data that passed the quality checks is left exactly as reported.

FOUR VERSIONS

It saves the data four ways — **unadjusted**, **outliers fixed**, **gaps filled**, and **both** — so you can always see exactly what changed.

How the fix works

When the previous module flags a month as an **outlier** or **missing**, FASTR replaces just that one value — using the facility's **own surrounding months**, never numbers borrowed from another facility. It works down a short ladder and takes the first option the facility's history allows:

1. **The months right around it** — the average of the months on either side of the flagged one. This is the normal case: the facility's own level at that point in time
2. **If the flagged month sits at the very start or end of the records** — there aren't enough months on *both* sides, so FASTR uses whichever side has data: the **6 months just after** (when the gap is near the beginning) or the **6 just before** (when it's near the end)
3. **For an outlier only — the same month a year earlier** — for seasonal services, this matches like with like (a December to a December)
4. **If none of those exist** — the facility's **overall average** for that indicator

Months that passed the quality checks are left exactly as reported, so the real shape of activity is preserved.

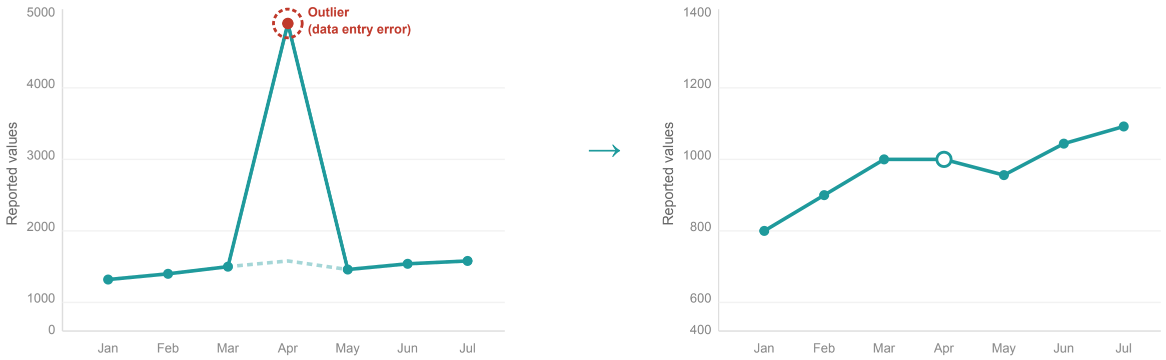
Some indicators are never adjusted: deaths and stillbirths (every case matters and must not be smoothed away), and very low-volume indicators — those that never reach 100 in a month, where there is too little signal to estimate from. These keep their raw values.

> ***Adjustment fills and smooths using each facility's own history — it never borrows from other facilities and never invents a trend.***

WHAT THE FIX DOES

A spike, replaced — the trend kept

Before adjustment | **After adjustment**



The spike compresses the Y-axis,
hiding meaningful variation in the data.

Outlier replaced with rolling average.
Underlying trend preserved.

Adjusting outliers improves the reliability of downstream utilization and coverage estimates

Each flagged value is swapped for the facility's own normal level, so the spike disappears but the real shape of activity stays intact.

THE OUTPUT

How much did the data change?

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausability of the data.

- **What each cell is** — pick a district (a row) and an indicator (a column). The number is how much that indicator's total **changed** once the suspicious spikes were removed. 0% = nothing needed fixing; a big number = big spikes were taken out
- **How to read it — green = the raw data was already clean; red = it needed a lot of correcting.** A red cell is a warning that the *raw* total there was inflated and would have overstated activity

> ***Worked example — Karene District → Family planning (long-acting): 5.7%, red** means fixing outliers cut that indicator's total by about 6% there; without the fix you'd have over-counted those services. A big correction isn't a failure — it's a sign the raw data would have misled you, and now won't.*

The platform also produces this table for **gaps filled** and for **both fixes together** — compare them to see whether spikes or missing months mattered more in an area.